

Faculty of Engineering & Technology

Electrical & Computer Engineering Department

Artificial Intelligence – ENCS3340

Second Semester (2025/2026)

Project 2: Machine Learning Classification Project

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Date:

JUNE 23, 2026

Abstract

This project applies machine learning classification techniques using Scikit-learn. It includes implementing a Decision Tree and a second classifier on the PlayTennis dataset, as well as evaluating a model on a UCI dataset. The models are compared using accuracy and evaluation metrics, showing differences in performance depending on the dataset.

1. Introduction

Machine Learning is a field of Artificial Intelligence in which machines are made to learn by themselves using data without explicit programming and making predictions out of it. Machine Learning has wide applicability in several real-life scenarios such as classification, prediction, and data analysis.

The main goal of this project is to utilize classification through Python programming language and Scikit-learn library. This project involves three phases in which the first phase will involve building a Decision Tree algorithm on the PlayTennis data set; second phase involves comparison between Decision Tree and any other classifier such as Naïve Bayes; third phase involves analyzing any other dataset obtained from the UCI Machine Learning Repository using the same classification algorithm with performance measures.

2. Part A: Decision Tree Classification

2.1 Dataset

The PlayTennis dataset will be utilized in this section of the project to implement the Decision Tree classification method. The dataset comprises 14 observations describing various weather conditions and predicts whether the tennis match will take place or not.

Four important features can be identified in this dataset, namely Outlook, Temperature, Humidity, and Wind. The target feature is PlayTennis with Yes/No as possible outputs.

This dataset assists in examining the connection between weather conditions and the probability of playing tennis, which essentially depends on weather factors.

2.2 Implementation

The categorical variables were processed with the help of OrdinalEncoder to convert all the features into numeric, while LabelEncoder was employed to encode the target variable for the application of the machine learning model.

Afterwards, a Decision Tree Classifier was trained with the use of the entropy criteria to learn the dependence of playing tennis from the characteristics of weather (attributes: Outlook, Temperature, Humidity, Wind). It was trained on the entire dataset to obtain rules for the classification.

Accuracy of the model was used for the evaluation of its performance; additionally, other parameters such as depth and leafs number of the tree were obtained to investigate the complexity of the model. The decision tree itself was graphed.

2.3 Results

The Decision Tree model achieved perfect performance on the PlayTennis dataset, reaching an accuracy of 100%. The tree depth was 4, and it contained 7 leaves, reflecting how the model splits the data into clear decision rules.

The results indicate that the model successfully learned the relationship between weather conditions and the decision to play tennis on the given dataset.

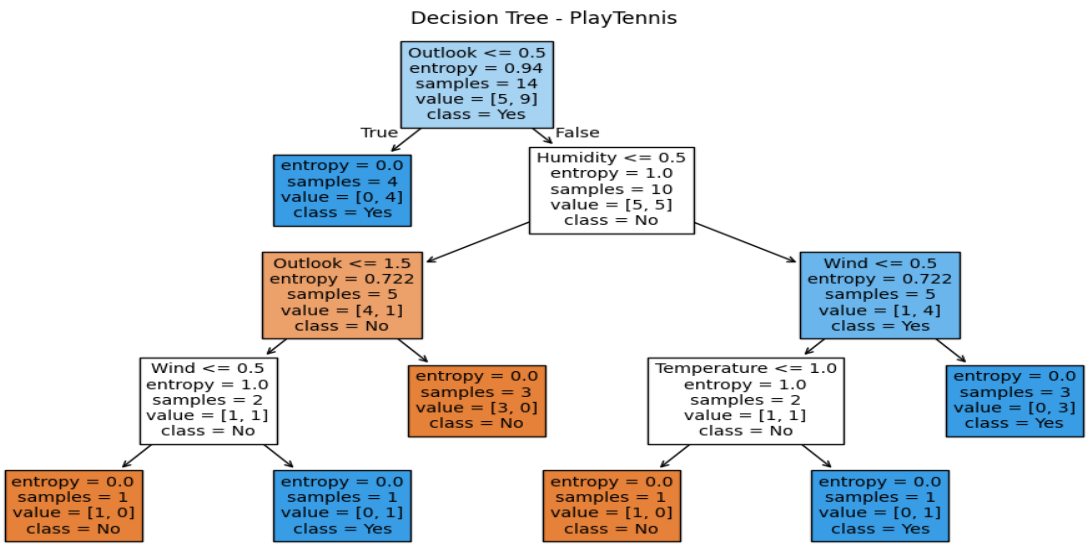


Figure 1:Decision Tree

The figure above shows the Decision Tree constructed for the PlayTennis dataset, where the data is split based on weather attributes such as humidity and outlook to reach the final decision.

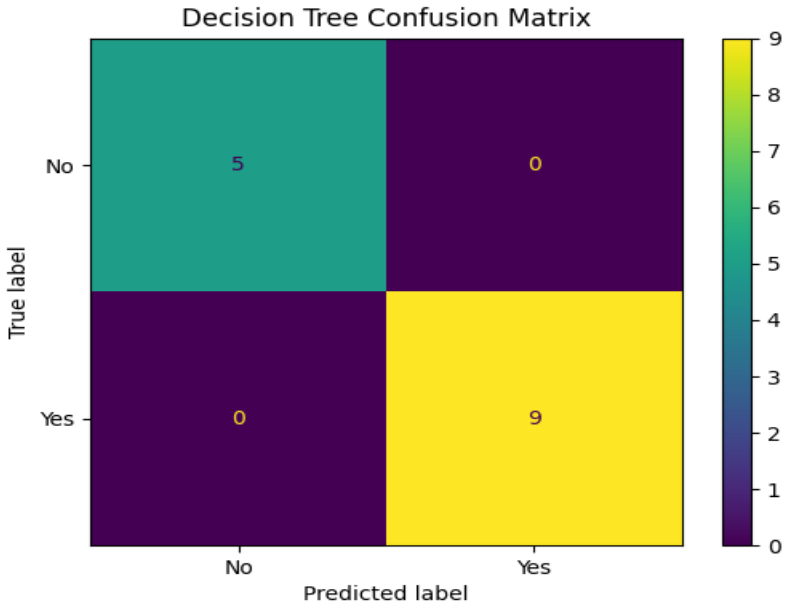


Figure 2: Decision Tree Confusion Matrix

The confusion matrix illustrates the performance of the Decision Tree model by comparing actual and predicted values, showing the number of correct and incorrect predictions.

2.4 Interpretation

These results reveal that the Decision Tree algorithm was able to effectively learn the relationship between weather data and the choice to go and play tennis. Among the attributes considered by the model, those with higher impact on the decision include Humidity and Outlook. It should also be noted that the model can be easily interpreted because of the rule set it forms.

3. Classifier Comparison (Naïve Bayes)

3.1 Dataset

The PlayTennis dataset used in part A was used in this part of the project. The dataset includes four major attributes: Outlook, Temperature, Humidity, and Wind together with the output PlayTennis. This is to ensure fairness in comparisons between different models.

3.2 Implementation

The Naïve Bayes (CategoricalNB) classifier was applied as a second model to compare with the Decision Tree. The model was trained on the same dataset after converting categorical values into numerical form using OrdinalEncoder and LabelEncoder.

3.3 Results

The Naïve Bayes model achieved an accuracy of 92.86%, which is lower than the Decision Tree model that achieved 100%. This shows that Decision Tree performed better on this dataset. A bar chart was used to visually compare both models.

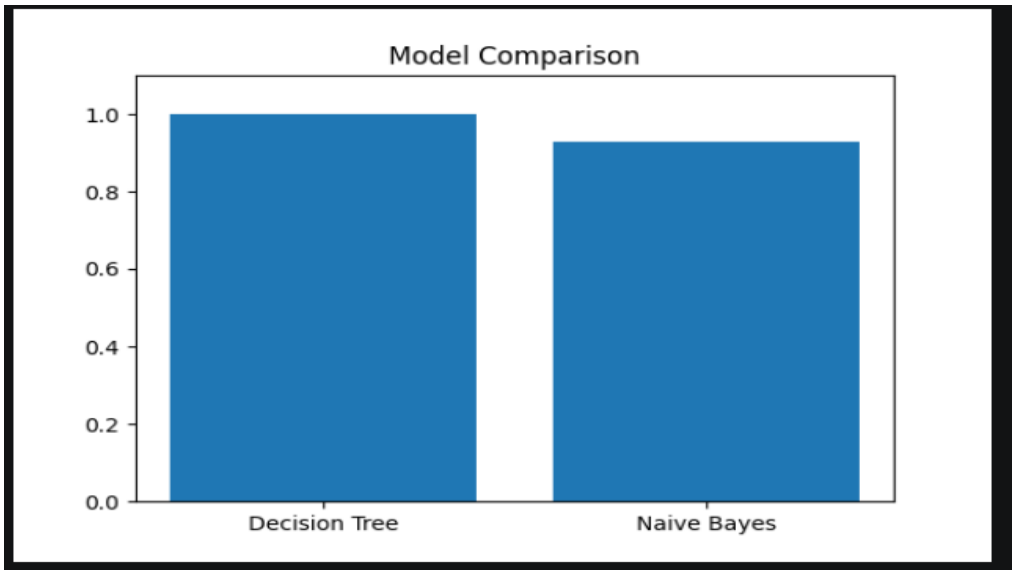


Figure 3:Comparison Between Decision Tree & Naiva Bayes

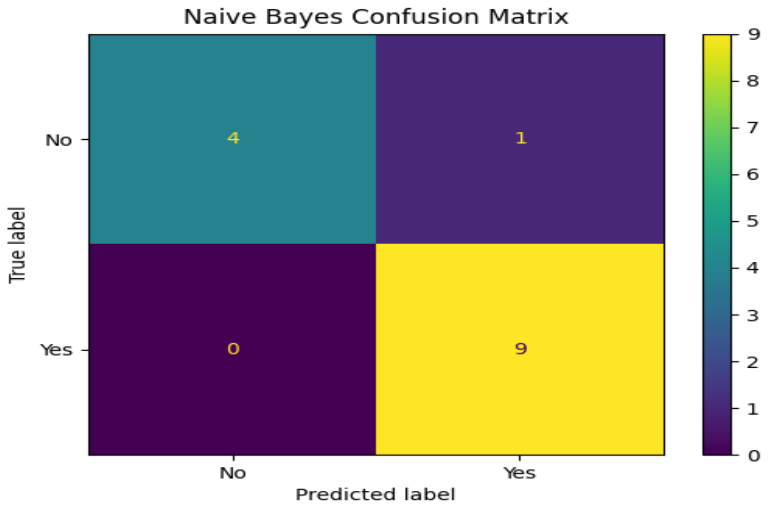


Figure 4:Naiva Bayes Confusion Matrix

The confusion matrix shows the classification performance of the Naïve Bayes model by displaying the correct and incorrect predictions.

3.4 Interpretation

The results show that each model uses a different learning approach. Decision Tree is rule-based, while Naïve Bayes is probability-based. In this case, Decision Tree achieved higher accuracy, while Naïve Bayes is simpler and faster.

4. Classification on UCI Dataset

4.1 Dataset

The Iris dataset was used in this part of the project. It contains 150 samples from three classes of Iris flowers: Setosa, Versicolor, and Virginica. The dataset includes four features: Sepal Length, Sepal Width, Petal Length, and Petal Width. The target variable represents the flower class, and the goal is to train a classification model to distinguish between these classes.

4.2 Implementation

The Iris dataset was loaded using Scikit-learn and split into training and testing sets with an 80/20 ratio. A Decision Tree classifier using the entropy criterion and maximum depth of 4 was trained on the training data. The model was evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

4.3 Results

The Decision Tree model achieved an accuracy of 96.67% on the Iris dataset. The classification report shows high performance across all classes, with strong Precision, Recall, and F1-score values. The confusion matrix also indicates that most predictions were correct, with only a few misclassifications between similar classes

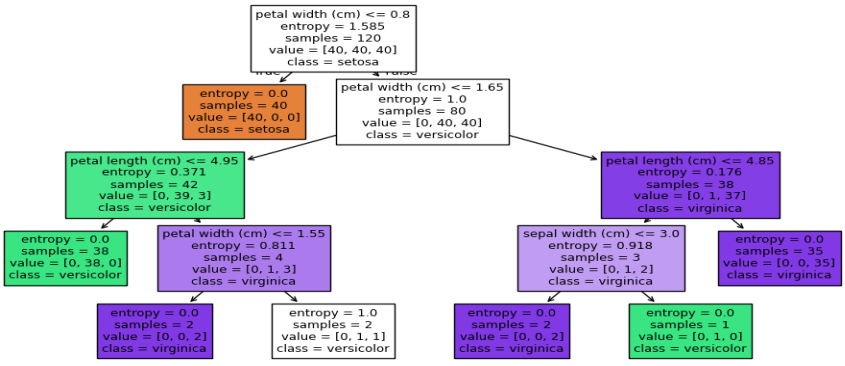


Figure 5:Decision tree Iris

The figure above shows the Decision Tree used for classifying the Iris dataset, where the data is split based on flower features such as Petal Length and Petal Width to reach the final class prediction.

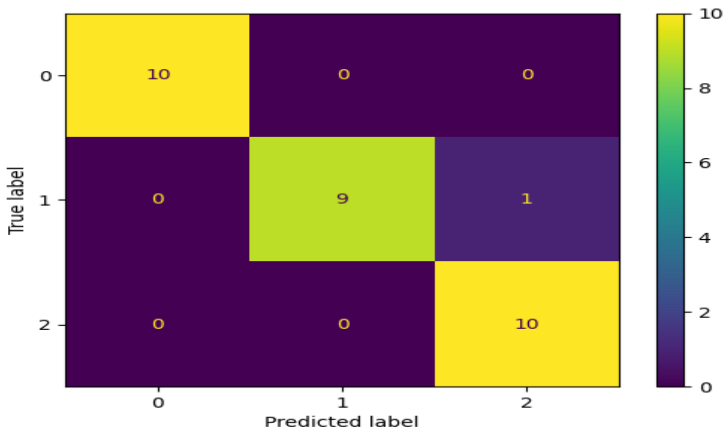


Figure 6:Confusion matrix illustrates

The confusion matrix illustrates the performance of the Decision Tree model by comparing actual and predicted values, showing correct and incorrect predictions for each Iris flower class.

4.4 Interpretation

The results show that the Decision Tree model performs very well on the Iris dataset, achieving high accuracy in distinguishing between the three flower classes. The model is able to identify important features such as Petal Length and Petal Width, which play a key role in separating the classes. Most errors occur between similar classes such as Versicolor and Virginica due to overlapping feature values.

5. Conclusion

This project applied different machine learning classification algorithms using Scikit-learn. A Decision Tree was implemented on the PlayTennis dataset and showed high performance in learning patterns from weather data. A second classifier (Naïve Bayes) was also applied and compared, where Decision Tree achieved higher accuracy.

In addition, the Iris dataset from the UCI repository was used, where the model achieved 96.67% accuracy, showing strong classification performance. Overall, the project demonstrates that different algorithms perform differently depending on the dataset, and model selection plays an important role in achieving good results.